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Tool and Fixture Design For Suppliers

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1 SCOPE:

1.1 This specification defines the process and requirements for the design and definition of Special to Product Tooling (STPT) for use on Rolls-Royce Components and Assemblies.

1.2 This specification defines the requirements for all Suppliers contracted to create tool and fixture designs for Rolls-Royce and is applicable to STPT used in the manufacture, assembly, repair and transportation of Rolls-Royce Components and Assemblies.

1.3 The following are out of scope of this specification:

- Make to print tooling
- Individual Proprietary and Standard parts or tools that do not form an integral part of tooling assemblies in Categories A and B
- Equipment that is designed specifically to lift Products, Components or other items.
- Special to Product Tools and Fixtures used in the Marine Business

2 REASON FOR USE:

2.1 To provide a global standard for the design of STPT that utilises Rolls-Royce's proven design and design review processes by providing a framework to ensure a common approach to tool design including methods, risk assessment, safety, usability, cost, deliverables and to ensure that Rolls-Royce requirements and other applicable directives and regulations are met.

3 APPLICABLE DOCUMENTS:

3.1 The following publications form a part of this document to the extent specified herein. The applicable issue shall be the current issue, unless a specific document issue is specified. When the referenced document has been cancelled and no superseding document has been specified, the last published issue of that document shall apply. Additionally, contact your sector's Rolls-Royce Technical Authority to obtain guidance/approval on action to remove or replace the cancelled reference by a drawing and/or specification change. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence.

3.2 ISO Documents: International Organization for Standardization, 1, Ch. de la Voie-Creuse, Case Postale 56, CH-1211 Geneva 20, Switzerland, or www.iso.org

ISO 14001 Environmental Management

3.3 Rolls-Royce documents:

The following are available internally from the Rolls-Royce Infocentre website http://www.infocentre.rolls-royce.com and Teamcenter and is available externally from https://suppliers.rolls-royce.com/

EDNS01000308516	AT&F User Guide – Conducting Design Reviews Applicable to Special to Product Tooling (STPT)
EDNS01000308520	AT&F User Guide – Design Guidance Applicable to Special to Product Tooling (STPT)

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EDNS01000308523	AT&F User Guide – Managing Risk Applicable to Special to Product Tooling (STPT)
EDNS01000308524	AT&F User Guide – Tool Drawing Guidance Applicable to Special to Product Tooling (STPT)
RRES 90057	Materials and Processes To Be Avoided

4 DEFINITION OF TERMS:

4.1 Throughout this document the words “tool” or “tooling” shall mean STPT including jig, tool, fixture, assembly tooling/jig, ground support equipment (GSE), repair tooling etc.

4.2 Associated definition of terms:

Assembly Engineering:
Any Rolls-Royce business function responsible for the assembly (build) of Components

Assembly Tooling:
Any ‘Special to Product’ equipment that is not an integral part of a Rolls-Royce product but used in the build, strip (disassembly), overhaul, protection, handling, transportation, installation, storage or testing of all or part of a Rolls-Royce product or any Component associated with the Rolls-Royce product.

Component:
The product of a manufacturing process and may comprise any part that is required to make the whole assembly. The Component is usually the item to which the STPT is designed to accommodate.

Definition:
In this context, the evolving description of all deliverables necessary to support production and operational support and disposal of a tooling package and its individual parts throughout its lifecycle

Drawing:
A Definition against which an item of tooling is procured and, in the case of a tooling assembly, its arrangement of constituent parts is defined.

Drawing Issue:
A particular update standard of any tool drawing and is identified with a letter which changes when the drawing is altered.

Fixture:
A piece of equipment used to position, hold and restrain a Component whilst a manufacturing or repair process is being performed.

Formal Communication
Communication whereby traceability is deemed necessary such as an official request for or transfer of information, where a decision has been made or an agreement has been reached.

Ground Support Equipment:
Storage & Transportation stands, containers, & associated accessories including turnover equipment, blanks, caps etc.

Jig:
A type of custom-made tool used to control the location and/or motion of another tool. A

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jig's primary purpose is to provide repeatability, accuracy and interchangeability during the manufacture or assembly of products.

Manufacturing Engineering:
The function within any Rolls-Royce business that is responsible for creating, implementing and maintaining processes used in the manufacture of Components or Component assemblies.

Product:
An item offered for sale and can be physical or virtual such as a service.

Proprietary Part:
The product of companies other than Rolls-Royce group which are not covered by any Rolls-Royce specification or controlling document, are not National or International standard parts, and are only available from that source

Risk:
An uncertain event (or set of circumstances) associated with an activity that, should the event occur, will have an impact on the achievement of one or more of the activity's objectives. Risk is thus measured in terms of the probability of an event and its impact on meeting the activity's objectives, such that it can be assessed and managed.

Responsible Rolls-Royce Business Function:
In the context of this specification, the function within Rolls-Royce that is responsible for acquiring the item(s) of tooling to facilitate the manufacture or assembly of Rolls-Royce components i.e. Manufacturing Engineering within a Supply Chain Unit (SCU)

Standard Part:
A part whose design definition is held in the public domain (e.g. Aerospace Defence and Security Industries (ADS), Society of Automotive Engineers (SAE), National Aerospace Standards Committee (NAS) part numbers, Deutsches Institut für Normung (DIN)'). These are parts that are used repetitively, may be obtained commercially and have their own separate identity.

Supplier:
An approved company delivering products or services to Rolls-Royce against a Rolls-Royce purchase order. This definition also includes the delivery of products and services between Rolls-Royce facilities.

Tool Designer:
The Tool Designer is the person responsible for designing and defining the tool or fixture to meet the requirements of the Tool Design Specification (TDS). The Tool Designer is expected to have a working knowledge of machine shop practices, tool making procedures, machine tool design, manufacturing procedures and methods, and possess conventional engineering disciplines such as planning, designing, engineering drawing and cost analysis. The Tool Designer can be either a Rolls-Royce employee or the employee of an external tool design Supplier.

Verification:
Confirmation, through the provision of objective evidence, that specified requirements have been fulfilled.

5 ACRONYMS:

ALARP As Low As Reasonably Practicable
DFMEA Design Failure Mode and Effects Analysis

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ECM	Engineering Co-ordination Memorandum
FEA	Finite Element Analysis
GSE	Ground Support Equipment
HS&E	Health, Safety & Environment
MFMEA	Machinery Failure Mode and Effects Analysis
MRO	Maintenance, Repair & Overhaul
NDA	Non-Disclosure Agreement
PFMEA	Process Failure Mode and Effects Analysis
PLM	Product Lifecycle Management
PoC	Point of Contact
RFQ	Request for Quote
STPT	Special to Product Tooling
TDR	Tool Design Review
TDS	Tool Design Specification

6 APPLICABILITY:

6.1 This specification applies to all Suppliers involved in the tool design process.

6.2 This specification is applicable to all Category A STPT used in the manufacture, assembly, repair and transportation of Rolls-Royce Components and Assemblies. In this context, Category A STPT is defined as:-

- Tools that fall under the Machinery Directive 2006/42/EC (implemented in the UK by the Supply of Machinery (Safety) regulations 2008).
- Tools where risk assessment results in the need for mitigation action such as incorporation of safety features, interlocks, guards etc.
- Tools where risk assessment results in the need for mitigation action to prevent detrimental impact on the facility or the environment.
- Tools where forces or torques are being applied either through mechanical, pneumatic or hydraulic pressure systems and where associated high values of stored energy exist.
- Tools which pose a high risk to the Component(s) such as loss of performance or reduction in life of a Component or Assembly.
- Tools which attach to, or may come into contact with, critical or highly stressed areas of Components or Assemblies including where operation of the tool is critical to Component or Assembly integrity such as the compressor lock plate bending tool which is critical to the final shape of the lock plate.
- Any equipment specifically, or as an integral additional function, designed to enable personnel to access areas at height.
- Tools that are complex and/or novel and require full design with innovation and/or requires intellectual property protection.

6.3 For Category B STPT, the requirements defined in sections 9, 11, 12 and 15 may be reduced or omitted as specified within each of those sections. In this context, Category B STPT is defined as:-

- Tools that contain low technical difficulty with minimal design content such as masking tools, process baskets, protectors etc.
- Tools where low or no forces are being applied such as STPT gauging, manual and CMM inspection fixtures etc.
- Tools which are predominantly based on existing tool designs within a well-established tooling family template where simple modifications are incorporated for such purposes as in-process improvements or as a result of manufacturing method changes etc. In this case, where lengthy service experience shows that the tool

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design on which the modification is being applied has not suffered any failure, has continually performed as required and will remain fit for purpose after the modifications have been incorporated, the responsible Rolls-Royce business function tooling technical authority shall decide that the resultant technical risk is acceptable such that Category B criteria can be applied. However, a HS&E assessment to the latest requirements shall be used to support the application of the Category B criteria but, if the HS&E assessment highlights risks that require mitigation, then the full requirements of this specification are mandated.

6.4 The responsible Rolls-Royce business function will specify the category of tool to be designed to the Supplier in the TDS (ref paragraph 7.4).

6.5 It is a mandatory requirement that the Supplier shall obtain approval for the Supplier's final design from the responsible Rolls-Royce business function requesting the tool design. This is usually achieved at the Critical Design Review (refer to paragraph 12.9).

7 REQUIREMENT CAPTURE:

7.1 It is the responsibility of the responsible Rolls-Royce business function to develop, capture and formally issue the requirements for every tool design in an approved format. The method by which these requirements will be delivered to the Supplier is by the creation of a Tool Design Specification (TDS).

7.2 The TDS is the document which communicates the technical requirements of the tool to the Supplier and will contain sufficient, accurate information so that the resulting tooling is delivered to programme timescales and is fit for purpose.

7.3 All TDS's shall specify that compliance to this specification is mandatory.

7.4 The TDS shall specify the category of the tool design to be undertaken (ref paragraph 6.2 and 6.3).

7.5 The TDS shall be initially issued with the Request for Quote (RFQ) and will contain all requirements that are available at the time of the initial issue.

7.6 Requirements may develop in stages to allow concurrent design activities and it is the responsibility of the responsible Rolls-Royce business function to keep the TDS up to date with the latest developments during the tool design process. However, the Supplier shall ensure that they have the latest version of the TDS at regular intervals during the tool design process and, in particular, before the Critical Design Review.

7.7 Tool design Suppliers shall provide input and support to the development of the requirements to ensure their satisfactory completion in line with planned and agreed arrangements. The Supplier shall determine:

- Requirements not stated by Rolls-Royce but necessary for specified or intended use, where known
- Applicable statutory and regulatory requirements not stated by Rolls-Royce
- Any additional requirements determined by the Supplier

7.8 The Supplier shall formally review all Rolls-Royce requirements specified in the TDS and shall:

- Ensure that the requirements are complete (or, where this isn't the case, that there is clear indication and rationale for this)

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- Ensure that the requirements are unambiguous, adequate, and not in conflict with each other
- Confirm that it has the capability to meet the requirements
- Confirm that it has the means of verifying compliance with the requirements
- Assess the maturity of the requirements and feed the findings back to Rolls-Royce
- Identify and communicate any issues or high-risk items with the requirements to the responsible Rolls-Royce business function in a timely manner

7.9 The addition of new requirements outside of the contractual scope or changes to the requirements after contract award shall be managed through the appropriate Rolls-Royce process.

7.10 If applicable, the Supplier shall flow down all applicable Rolls-Royce Technical Requirements (including regulatory) into their own supply chain as part of maintaining demonstrable compliance to the requirements.

7.11 The Supplier shall ensure its Tool Design outputs comply with all Rolls-Royce requirements and shall:

- Maintain mechanisms to demonstrate complete requirements traceability (refer to paragraph 7.6).
- Maintain mechanisms for the incorporation of new/updated requirements (refer to paragraph 7.6).

8 TOOL IDENTIFICATION:

8.1 It is the responsibility of the responsible Rolls-Royce business function to provide the necessary tool identification numbers to the Supplier.

8.2 All tooling and fixturing shall be identified with a unique number.

8.3 All relevant designs, drawings, specifications, reports, documents communications etc. shall specify this number to ensure traceability.

9 PROJECT MANAGEMENT:

9.1 Tool delivery can have a significant impact on Rolls-Royce programmes in terms of product delivery to the customer and, in order to manage this, a bespoke tool design or design/make plan shall be created to ensure the Rolls-Royce product is delivered as committed. This bespoke tool design plan can take the form of a very simple list containing key milestones to a detailed project plan.

9.2 It is the responsibility of the Supplier to create and own this plan and to ensure adequate progress is being made such that the tool will be delivered according to the agreed plan.

9.3 The Supplier shall engage with Rolls-Royce in the development and progress monitoring of this plan and gain agreement from Rolls-Royce to the commitment dates within the plan.

9.4 The Supplier shall plan the tool design activities in collaboration with the responsible Rolls-Royce business function PoC but consultation with all relevant stakeholders shall take place to ensure delivery of the tool meets the delivery requirements of the

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Rolls-Royce product. The stakeholders can include, but are not limited to, Rolls-Royce Manufacturing Engineering, Rolls-Royce Integrated Programme Team Leader, Rolls-Royce Design Engineering, Rolls-Royce Assembly Engineering and the Tool Manufacturer. The final plan shall have the agreement of all the key stakeholders to ensure timescales and resource commitments can be met.

9.5 As a minimum, the bespoke tool design plan shall contain dates for the following milestones:

- Commencement of the tool design
- Tool Design Reviews (refer to section 12)
- Risk assessments (MFMEA's, initial and subsequent) - (refer to section 11)
- Rolls-Royce key input requirement dates - (refer to paragraph 9.7)
- Tool drawing final issue

9.6 In the case of tool design/make programmes, the bespoke tool design/make plan shall include tool manufacturing milestones including commencement of tool manufacture and delivery of the tool to Rolls-Royce.

9.7 To support concurrent engineering practices and reduce timescales, the Supplier shall consider when key information regarding the Rolls-Royce product is required to be delivered by Rolls-Royce in order for the Tool Design to meet the Rolls-Royce product delivery commitments. Key information in this context could be Component features that interface or interact with the tool such as tooling location features, datums, other significant features, access limits etc. The Supplier, in conjunction with the responsible Rolls-Royce business function PoC, shall then negotiate with the Rolls-Royce product Designer to agree the dates for these features to be fixed within acceptable physical limits and provided to the Supplier. This information shall be formally captured in stages during the Component design phase using the communications process described in section 10.

9.8 For Category B tool designs, the minimum requirements defined in paragraph 9.5 may be waived at the discretion of the responsible Rolls-Royce business function. In this case, the final issue date alone shall be agreed.

10 COMMUNICATION:

10.1 Communication is an integral part of the tool design process and it is the Supplier's responsibility to ensure that all relevant information is available to enable the tool design to progress. The Supplier is responsible for pro-actively seeking out any information that is required and to escalate issues to the responsible Rolls-Royce business function point of contact (PoC).

10.2 In order to maintain an adequate level of traceability, all formal communications, including those between the Supplier and Rolls-Royce shall:

- Be dated
- Be uniquely identified
- Carry applicable part numbers where appropriate

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- Identify the correspondence initiator
- Be sent/received through identified points of contact
- Specify targets for communication responses.

10.3 The Supplier may be requested by the responsible Rolls-Royce business function to use the standard Rolls-Royce Engineering Co-ordination Memorandum (ECM) process for this purpose. If this is the case, the Supplier shall comply with Rolls-Royce methods of controlling and using ECM's.

11 RISK MANAGEMENT:

11.1 All designs, whether a new or a scaled version of a previous design, carry a level of risk even if the perception is that it is small. Along with requirements, risk shall be reviewed periodically throughout the project with continuous updates as potential risks are managed and treated.

11.2 Based on this requirement all new tool designs shall be subject of a Machinery Failure Mode and Effects Analysis (MFMEA). MFMEA is a technique used throughout industry for evaluating equipment and tooling during its design phase to improve operator safety and the reliability and robustness of the tooling or machinery.

11.3 The Supplier shall therefore perform the MFMEA on their tool designs as defined in this specification and shall collaborate with Rolls-Royce to:

- Define risk assessment criteria and ratings
- Identify, record, prioritise, track, and mitigate technical and HS&E risks
- Assign owners and resolve risk mitigation actions
- Regularly communicate risk assessments and mitigation progress to the responsible Rolls-Royce business function

11.4 The Supplier shall also:

- Demonstrate an appropriate design optimisation process, in which findings from the MFMEA are fed back into the design process.
- Review and update its MFMEA in response to any requirements change and provide Rolls-Royce with the update.

11.5 All Risks identified by the MFMEA, shall be recorded, tracked, and mitigated through inclusion in the Supplier's Risk Management System.

11.6 MFMEA is used as the means of technical risk management within the tooling and equipment design process, however, the MFMEA shall also consider Health, Safety & Environment (HS&E) risks.

11.7 As well as de-risking the chosen tool design to a level that is "As Low As Reasonably Practicable (ALARP)", the outcome of the risk assessment will help with the down-selection of the concepts being considered as some may contain risks that make a particular concept unviable. The MFMEA process shall therefore be initiated as early as the concept design phase (refer to paragraph 12.9).

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11.8 Existing MFMEA's shall be re-used when the tool design is part of a particular "tooling family" where the purpose and features of the design are the same or similar such that the new MFMEA will focus only on the new or novel features of the new tool design. Where this is the case, the responsible Rolls-Royce business function shall provide any existing completed MFMEA's to the Supplier. Where no MFMEA exists, the Supplier shall create a new one.

11.9 As well as considering how failure of the individual key features of the tool will impact the ability to meet the specific requirements defined in the TDS, the MFMEA shall also consider, as a minimum, how the failure of the individual key features of the tool will:

- Injure the operator (both installing the tool as well as operating it)
- Damage the Component or surrounding Components
- Damage equipment, machinery, surroundings or the environment
- Impact the ability to manufacture or assemble/disassemble the Component(s)

11.10 The Supplier may be requested to consider certain Component DFMEA and PFMEA risks that the tool's design may be able to influence. These risks will be provided by the responsible Rolls-Royce business function and included in the TDS. These additional risks shall be captured in the tool MFMEA to show how the Tool's Design mitigates or avoids these relevant risks.

11.11 The MFMEA is a live document and will develop as the tool design progresses and, as such, it shall be updated and then reviewed at the relevant Design Reviews (refer to section 12). The Supplier will be expected to answer questions at the Tool Design Reviews to confirm that MFMEA is being conducted and identified risks treated appropriately. At each review, it is a requirement that the MFMEA has been conducted to a sufficient level to identify risks and mitigations as the tool design progresses towards completion.

11.12 It is the Supplier's responsibility to plan, conduct and own the tooling MFMEA process with support from the responsible Rolls-Royce business function subject matter experts.

11.13 If necessary, further information regarding tool design risk management, criteria and ratings, including a STPT MFMEA template that can be copied and modified, can be found in EDNS01000308523 - AT&F User Guide – Managing Risk Applicable to Special to Product Tooling (STPT)

11.14 The requirement to conduct a MFMEA on Category B tool designs, whilst recommended, is at the discretion of the responsible Rolls-Royce business function tooling technical authority.

12 TOOL DESIGN REVIEW PROCESS:

12.1 The purpose of the Tool Design Review process is to gain and formally record technical approval to release new or changed definition from an appropriately authorised person. Technical approval signifies that the definition will provide an acceptable solution to the requirements. The procedure operates as required ahead of definition release points. As a result, all new tool designs shall be subject to Tool Design Review (TDR).

12.2 Tool Design Reviews shall be conducted by a reviewer or review team from the responsible Rolls-Royce business function.

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12.3 The Supplier shall actively participate in the Rolls-Royce Tool Design Review process as defined in this specification and shall:

- Agree to the applicability of each Tool Design Review gate in relation to the scope of the tool design.
- Agree the Tool Design Review schedule with Rolls-Royce.
- Agree with Rolls-Royce the roles, accountabilities, responsibilities, deliverables, attendees, and agenda for each Tool Design Review.
- Attend Tool Design Reviews in accordance with planned and agreed arrangements.

12.4 The Supplier shall plan their work to facilitate progression through this Tool Design Review process. The Supplier shall own and update the Tool Design Review plan, record actions that result from the reviews, including action owners, and update the review panel on the status of the actions between reviews.

12.5 The Supplier shall use the Tool Design Review Process shown in Figure 1 below when planning and participating in Tool Design Reviews.

12.6 Rolls-Royce will collaborate with the Supplier at the Tool Design Review, to evaluate the status of the design and identify any problems or risks, together with their necessary actions.

12.7 The Supplier shall keep a formal record of all minutes, actions and salient points agreed upon at each Tool Design Review and ensure effective and timely close-out. Ideally, this will be via a copy of the completed checklist used at the Tool Design Review.

12.8 Rolls-Royce will give a clear indication of the result of the Tool Design Review and ensure the result is discussed and agreed prior to the close of the meeting.

12.9 There are principally four stages to the Tool Design Review process starting with Project Launch and culminating in the Critical Design Review. It is at the discretion of the responsible Rolls-Royce business function whether some or all of these reviews are formal or informal. The full Tool Design Review process shall be followed for Category A tool designs but reviews may be combined where appropriate. For Category B tool designs, the process is configurable such that a reduced process can be followed whereby the responsible Rolls-Royce business function shall select the reviews to be used taking into account the complexity of the tool and the risks associated with the tool design:

- Project Launch - The Project Launch Review is the “kick-off” meeting that initiates the Tool Design process. It is the key review with respect to requirements capture and sets the scene for the Tool Designer’s tasks.
- Concept Design Review - This review is where the tool design concepts are reviewed and their individual attributes with respect to achieving the requirements safely, within cost and on time are assessed.
- Preliminary Design Review – This review is where the chosen concept is reviewed with an explanation as to why this concept has been selected and why other potential concepts have not been selected.
- Critical Design Review - This review ensures that the design meets the requirements of the customer, is safe to use and all risks associated with the tool design have been

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mitigated to an ALARP level. The Critical Design Review is the final check that all tasks associated with the tool design have been completed and the tool design process has come to an end.

12.10 Detailed checklists shall be used by the reviewer(s) and Supplier to ensure that work is completed to an acceptable standard. As every Manufacturing and Assembly function within each Rolls-Royce business has their own tooling requirements, each function operating the Design Review Process will develop a list of pertinent questions applicable to each review that addresses the relevant topics including any questions that are specific to the area where the tool is to be used. These lists shall be used by the Supplier at each review to ensure the design meets all the necessary criteria in order to meet the tooling requirements. This will also form the official record of progress through the Tool Design Review process. For Category B tool designs, the use of checklists can be waived at the discretion of the responsible Rolls-Royce business function tooling technical authority in preference for informal discussions using information provided by the Tool Designer such as drawings, models etc.

12.11 Information that the Tool Design Reviewer(s) should be looking for at each review and what the Supplier is expected to have prepared for each review in order to gain agreement from the reviewer(s) that the design can progress on to the next stage can be found in EDNS0100030851 6 - AT&F User Guide – Conducting Design Reviews Applicable to Special to Product Tooling (STPT).

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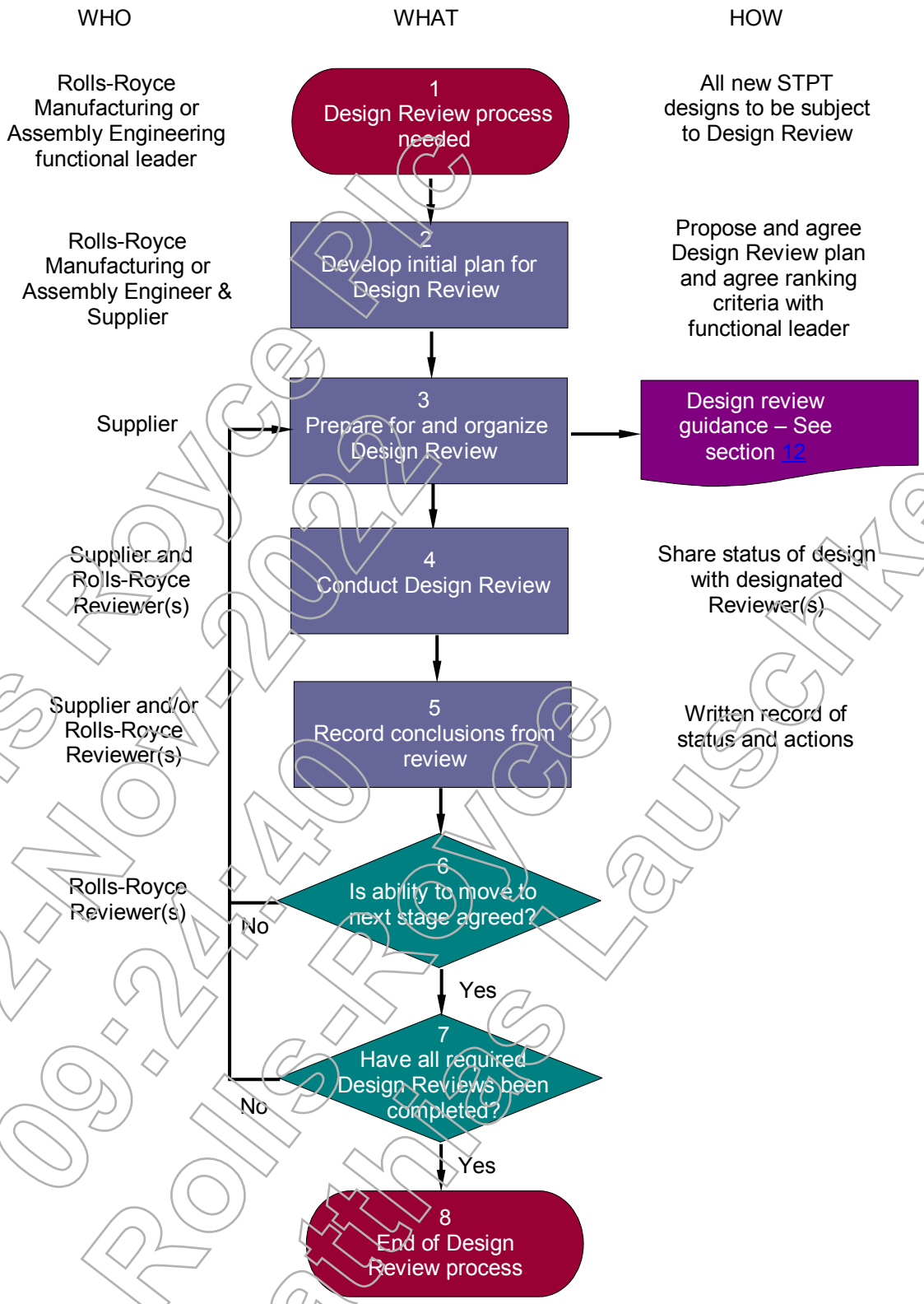

FIGURE 1 - TOOL DESIGN REVIEW PROCESS

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13 DESIGN:

13.1 The Supplier shall make full use of appropriate design methodologies and techniques and comply with the appropriate regulations throughout their design activities. These shall include, but are not limited to:

- Compliance with the relevant specifications, standards and Health, Safety and Environment (HS&E) requirements.
- Compliance with all applicable laws, codes, regulations, directives and legislations of the countries, states, and localities in which the tool is designed and operated.
- Use of appropriate design disciplines, such as analysis, including Stress and FEA (including Static and Dynamic analysis), process modelling, calculations, design within experience where possible, embedding of knowledge, lessons learnt and best practice from previous tool designs, MFMEA etc.
- Design for Manufacture, Assembly and Cost
- Design for Reliability
- Design for the prevention of foreign objects and contamination being trapped during manufacture, maintenance or repair in such features as, but not limited to, apertures, orifices, sharp bends and corners.
- Materials chosen shall be compatible with all current applicable Health and Safety standards including Environmental Standard ISO 14001. Certain materials and processes can be detrimental to the operation of engine components therefore; materials listed in Table 1 of RRES 90057 shall not be used in any tool or fixture without the permission of the responsible Rolls-Royce business function.

13.2 The tool shall be defined physically, functionally and operationally in accordance with requirements stated in the TDS. Any failure to meet the requirements shall be discussed and agreed at the relevant Tool Design Review.

13.3 Further help and guidance on the fundamentals of Tool design, including relevant specifications and standards, considerations and analysis techniques, can be found in EDNS01000308520 - AT&F User Guide – Design Guidance Applicable to Special to Product Tooling (STPT).

14 DEFINITION:

14.1 The tool drawing is the single authorised source of the "what to make" aspect of the tool definition and shall be identified by its part number as soon as one is available (refer to Section 8).

14.2 Model and drawing creation shall comply with agreed Rolls-Royce requirements.

14.3 Guidance on drawing presentation applicable to STPT can be found in EDNS01000308524 - AT&F User Guide – Tool Drawing Guidance Applicable to Special to Product Tooling (STPT).

15 DELIVERABLES AND STORAGE:

15.1 Each tool design will result in a standard set of deliverables. This provides traceability and a point of reference for future tool designs. The deliverables specified below are in

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relation to this tool design specification only and forms part of the total set of deliverables that each tool shall have when it is delivered.

15.2 Supplier Deliverables:

15.2.1 Models & Drawings:

- Model and drawing creation and storage in Teamcenter Engineering shall comply with agreed Rolls-Royce requirements.

15.2.2 Completed MFMEA:

- Where a MFMEA has been conducted, the completed MFMEA shall be provided with the design to provide evidence that all risks are mitigated to an ALARP level.

15.2.3 Analysis models and results:

- Any analysis models, results and calculations created to support the tool design shall be provided.

15.2.4 Completed TDS:

- An up to date copy of the TDS that the Supplier has been working to shall be supplied with the tool design.

15.2.5 User Guide:

- A user guide shall be provided to ensure that the operator knows how to use the tool safely.
- Any clamping features that require the operator to tighten using threaded fasteners to enable the Component to be secured into the tool shall have the torque value specified to ensure that the Component does not break free of the tool and to ensure that the clamp is not over-torqued such that excessive Component distortion or damage occurs.
- Any tool lifting requirements as well as any special operating instructions, clamping or installation sequence or safety precautions (particularly when hydraulic or pneumatic devices are incorporated into the tool design) shall also be included.
- The format of this user guide shall be agreed by the responsible Rolls-Royce business function at the Project Launch Review i.e. formal report template, on the drawing, directly into the Fixture Passport if available etc.
- Where necessary, maintenance requirements, critical inspections and the provision of 'life limits' (number of uses or service life) for tools or parts of tools shall be considered and included in the appropriate documentation accompanying the tool design.

15.2.6 Formal Communications:

- Copies of all formal communications shall be supplied to provide traceability of the pedigree of the design and provide a record of significant decisions made. It is proposed that these be embedded into an appropriate spreadsheet such that the whole document can be uploaded into Teamcenter rather than each individual communication.

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15.2.7 Plans:

- Up to date plans shall be supplied in order to facilitate programme management activities, particularly during the tooling manufacture stage.

15.2.8 Lessons Learnt:

- Any lessons learnt shall be supplied for future use.

16 CHANGE CONTROL:

16.1 Alterations to formally issued tooling drawings shall be controlled such that traceability is maintained.

16.2 Any change to the drawing **will** update the issue status.

16.3 The issue standard of a drawing shall be identified in alphabetical order; the first official issue of a tooling drawing will be issue A and all following issues will be identified sequentially omitting the letters I, O, Q, S, X and Z.

16.4 The issue letter shall be recorded on the body of the drawing in the appropriate location as dictated by the approved drawing format being used and details of the change captured in detail in accordance with local Rolls-Royce procedures.

16.5 Where the PLM toolset is being used, change control of the drawing within Teamcenter shall be in accordance with agreed Rolls-Royce requirements.

16.6 Any alteration to a tool definition shall ensure that current tools remain usable both before and after alteration.

16.7 The general principle employed is that a change to a formally issued tool drawing shall be acceptable provided that it does not adversely affect the safety, performance or function of the tool or parts of the tool concerned, outlined as follows:

- Always use stock
  - Existing tools shall continue to be usable without restriction. Alterations to tooling which will render it unsuitable for the original function will not have the drawing changed by alteration and, in this case, a new drawing shall be created under a new number.
- Scrapping of pre-alteration parts
  - Scrapping of the pre-alteration tools or parts of tools is not permitted.
- Manufacturability is not adversely affected
  - Changes to tolerances shall not render existing Tools non-conforming to the modified drawing specification.
  - Existing parts of a tool assembly shall not be removed by alteration but alternative parts may be added such that the assembly may be constructed from alternative optional parts.
- Definition errors/omissions

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- Shall be judged on their effect but will be acceptable if the part of the tool cannot be manufactured e.g. feature shown but dimensions omitted.

- Safety or reliability of the tool is not adversely affected
  - Safety or durability loss is not acceptable. Consistent and repeatable tool mechanical integrity shall be maintained.
- Material or process change
  - Existing materials or processes specified on the drawing shall not be removed but alternative material or process may be added provided the original material or process does not infringe the current Company Health, Safety and Environment policy and is not restricted by national laws, international conventions or protocols.
- Tools that have common usage
  - Due consideration shall be given to alterations to tool drawings that are used in other Rolls-Royce and non-Rolls-Royce facilities, for example Maintenance, Repair and Overhaul (MRO) facilities and approval is required from all relevant functions.

17 KNOWLEDGE CAPTURE:

17.1 A lesson's learnt log will be kept and maintained by each Rolls-Royce business function such that Tool Designers can review previous lessons and ensure that new tool designs embed these lessons.

17.2 The lessons learnt relevant to the tooling family will be referenced in the TDS and be provided to the Supplier for review. The Supplier shall embed all relevant lessons learnt into the new Tool Design.

17.3 It is the Supplier's responsibility to capture any new lessons, which can include such things as health & safety issues encountered, best practices developed, novel solutions to design problems, new analyses etc. and provide them to the responsible Rolls-Royce business function such that they can be recorded in the appropriate Rolls-Royce lessons learnt log.

18 INTELLECTUAL PROPERTY:

18.1 The Supplier shall not incorporate any export controlled items (tangible or intangible) into designs without informing Rolls-Royce and gaining prior written authorisation from Rolls-Royce.

18.2 The Supplier shall provide protection of Rolls-Royce Intellectual Property in its custody (including any third-party Intellectual Property provided by Rolls-Royce for the purposes of the contract) and prevent unauthorised disclosure to third parties. The Supplier shall:

- Put in place a Non-Disclosure Agreement (NDA) at the start of collaboration with Rolls-Royce and any third parties who need to be party to the collaboration.
- Prohibit the transfer or copy to a third party (including Supplier employees without the need to know) of Rolls-Royce Intellectual Property (including any third-party Intellectual Property provided by Rolls-Royce for the purpose of the contract collaboration) without the express written consent of Rolls-Royce.

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- Secure Rolls-Royce Intellectual Property (including any third-party Intellectual Property provided by Rolls-Royce for the purpose of the contract collaboration) from theft, fire, or other physical loss.
- Return or destroy data supplied by Rolls-Royce at the end of the contract or collaboration.
- Determine ownership of any jointly generated Intellectual Property.

18.3 The Supplier shall, upon request, give Rolls-Royce full access to details of the design for the purposes of Design Review, audit, and failure investigation.

18.4 If Rolls-Royce has funded in whole or in part the development of the technology used by the Supplier in the realisation of the design, then the Supplier shall provide Rolls-Royce with full details of the design.

18.5 If Rolls-Royce or the Supplier has funded a realisation of the design, based on technology that is the intellectual property of the Supplier, the Supplier shall provide Rolls-Royce with sufficient information to allow the tool to be integrated into the Rolls-Royce system.

18.6 Contractual obligations may take precedence over the intellectual property requirements outlined above. Rolls-Royce will negotiate any proposed deviation from their contractual obligations.

19 NOTES:

19.1 None.

Issue	Change (ESC) Number	Change Summary (In Descending Order)	Date
Initial	RFS 1015	EDNS04000046978	16 Jun 2015

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